

LEAN STATEMENTS FOR THE STATIONARY SOBOLEV MONOTONICITY FORMALIZATION

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1. CONVENTIONS

This document is a theorem-and-definition ledger for the Lean development. Proofs are intentionally omitted. Each theorem, proposition, lemma, or definition heading records the source Lean file and the Lean declaration name.

Lean's `Domain n` is read as $\mathbb{R}^n$, `Target m` as $\mathbb{R}^m$, and `Gradient n m` as the space of arrays $G = (G_i)_{i=1}^n$ with $G_i \in \mathbb{R}^m$. If $G : \mathbb{R}^n \rightarrow (\mathbb{R}^m)^n$, then

$$\Theta(G, a, \rho) := \rho^{2-n} \int_{B(a, \rho)} |G(x)|^2 dx$$

denotes the mathematical version of Lean's `weakTheta G a rho`. The annular radial-energy term

$$\mathcal{R}(G, a; s, r) := 2 \int_{B(a, r) \setminus B(a, s)} |x - a|^{2-n} |\partial_{r,a} G(x)|^2 dx$$

denotes the mathematical version of Lean's `weakMonotonicityRhs G a s r`.

2. MAIN PUBLIC THEOREMS

Theorem 2.1 (`MainTheorem.lean: stationaryW12LocMonotonicityFormula_euclidean`). *Let $n, m \in \mathbb{N}$ with $n \neq 0$. Let $\Omega \subset \mathbb{R}^n$ be a measurable set, let $u : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and let*

$$\mathcal{U} : \text{StationaryW12LocMap } u \Omega$$

be a stationary local $W^{1,2}$ witness with displayed weak gradient

$$G_{\mathcal{U}} := \mathcal{U}.w12.weakGrad.$$

Assume that $a \in \mathbb{R}^n$, $R_0, s, r \in \mathbb{R}$ satisfy

$$\overline{B(a, R_0)} \subset \Omega, \quad 0 < s, \quad s < r, \quad r < R_0.$$

Then the normalized weak-energy increment is exactly the annular radial-energy term:

$$\Theta(G_{\mathcal{U}}, a, r) - \Theta(G_{\mathcal{U}}, a, s) = \mathcal{R}(G_{\mathcal{U}}, a; s, r).$$

Theorem 2.2 (`MainTheorem.lean: stationaryW12LocMonotonicity_euclidean`). *Let $n, m \in \mathbb{N}$ with $n \neq 0$. Let $\Omega \subset \mathbb{R}^n$ be a measurable set, let $u : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and let*

$$\mathcal{U} : \text{StationaryW12LocMap } u \Omega$$

be a stationary local $W^{1,2}$ witness with displayed weak gradient

$$G_{\mathcal{U}} := \mathcal{U}.w12.weakGrad.$$

Assume that $a \in \mathbb{R}^n$, $R_0, s, r \in \mathbb{R}$ satisfy

$$\overline{B(a, R_0)} \subset \Omega, \quad 0 < s, \quad s \leq r, \quad r < R_0.$$

Then the normalized weak energy is monotone in the radius:

$$\Theta(G_U, a, s) \leq \Theta(G_U, a, r).$$

Remark 2.3 (`MainTheorem.lean`: theorem inputs). The Lean package `StationaryW12LocMap u Omega` contains the local L^2 data for u , the displayed weak gradient, local L^2 and a.e. measurability data for that gradient, the distributional weak-gradient identity, and the vanishing domain first-variation identity. The theorem does not assume any target-manifold constraint.

3. PLANNED FILE-BY-FILE TRANSCRIPTION

The remaining entries will be inserted one Lean file at a time. Within each file section, every public definition and every proved declaration (`def`, `abbrev`, `structure`, `theorem`, `lemma`, `instance`, and related named conclusions) will receive a corresponding LaTeX entry, with no proof text.

3.1. Public API layer.

StationaryHarmonicMap/Basic.lean.

StationaryHarmonicMap/Euclidean.lean.

StationaryHarmonicMap/WeakStationarity.lean.

StationaryHarmonicMap/WeakGradientBridge.lean.

StationaryHarmonicMap/L2LocBridge.lean.

StationaryHarmonicMap/SobolevBridge.lean.

StationaryHarmonicMap/FirstVariationBridge.lean.

StationaryHarmonicMap/StationarityBridge.lean.

StationaryHarmonicMap/SobolevWitness.lean.

StationaryHarmonicMap/StationaryMap.lean.

StationaryHarmonicMap/MainTheorem.lean.

StationaryHarmonicMap/API.lean.

Examples/UseMainTheorem.lean.

Examples/StationaryMonotonicity.lean.

3.2. Radial geometry and cutoff layer.

StationaryHarmonicMap/EnergyQuantities.lean.

StationaryHarmonicMap/RadialGeometry.lean.

StationaryHarmonicMap/RadialCutoffs.lean.

StationaryHarmonicMap/PrimitiveCutoffs.lean.

StationaryHarmonicMap/RadialIntegrability.lean.

StationaryHarmonicMap/RadialIdentity.lean.

3.3. Radial measure and radius calculus layer.

StationaryHarmonicMap/RadialMeasure.lean.

StationaryHarmonicMap/RadiusWeights.lean.

StationaryHarmonicMap/RadiusPrimitive.lean.

StationaryHarmonicMap/BallIntegralAC.lean.

StationaryHarmonicMap/RadiusWeightedDerivative.lean.

StationaryHarmonicMap/RadiusWeightedRepresentation.lean.

StationaryHarmonicMap/RadiusAnalysis.lean.

StationaryHarmonicMap/RadiusFormulas.lean.

3.4. Boundary identity and monotonicity layer.

StationaryHarmonicMap/BoundaryBasics.lean.

StationaryHarmonicMap/BoundaryFromRadial.lean.

StationaryHarmonicMap/MonotonicityFinal.lean.

StationaryHarmonicMap/MonotonicityRoutes.lean.

StationaryHarmonicMap/MonotonicityEuclidean.lean.

StationaryHarmonicMap/CenterTranslation.lean.

StationaryHarmonicMap/Monotonicity.lean.